Fleet mix planning

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GRA4154 Supply Chain Optimization with Mathematical Programming
Lecture 6

March 6, 2026

Contents

1 Problem

- A company manages the distribution of products from a factory to 4 customers
- The company needs to decide its fleet mix for the year, that is, which ships to hire
- There are 4 available ships to choose from:
  1. 2,000 tons, 8,1 MEUR
  2. 3,000 tons, 11.2 MEUR
  3. 4,000 tons, 14.4 MEUR
  4. 5,000 tons, 17.2 MEUR

There is 1 factory, and 4 customer sites with the following demands:

Site	**Demand**
1	70,000
2	90,000
3	100,000
4	60,000

- The company has created 6 possible routes for the ships
- Each route can have at most one ship assigned to it
- Each ship can be assigned to at most one route

1. Route 1 (23 days): Factory → 1 → 2 → 3 → 4 → Factory
2. Route 2 (18 days): Factory → 1 → 3 → 2 → Factory
3. Route 3 (13 days): Factory → 1 → 2 → Factory
4. Route 4 (13 days): Factory → 2 → 3 → 4 → Factory
5. Route 5 (15 days): Factory → 3 → 2 → Factory
6. Route 6 (20 days): Factory → 1 → 3 → 4 → Factory

1.1 Parameters

- loadcap_s: max load for ship s
- cost_s: yearly cost for ship s
- D_j: Yearly demand for customer j
- time_r: duration of route r in days
- $a_{r,j}$: cover matrix = 1 if customer j is in route r, else 0

1.2 Variables

- Y_s: binary variable, 1 if ship s is chosen, else 0
- $W_{s,r}$: binary variable, 1 if ship s is assigned to route r, else 0
- $Q_{s,r,j}$: yearly quantity delivered by ship s on route r to customer j
- AnnualCap_s: yealy delivery capacity of ship s, depending on assigned route

1.3 Model

$$\min Z = \sum_s \text{cost}_s Y_s$$

$$\text{s.t.}$$

$$\sum_r W_{s,r} \leq 1 \quad \forall\, s \tag{1}$$

$$\sum_s W_{s,r} \leq 1 \quad \forall\, r \tag{2}$$

$$Y_s \geq W_{s,r} \quad \forall \text{ combinations of } s \text{ and } r \tag{3}$$

$$\sum_s \sum_r Q_{s,r,j} = D_j \quad \forall\, j \tag{4}$$

$$Q_{s,r,j} \leq a_{r,j} D_j W_{s,r} \quad \forall \text{ combinations of } s, r \text{ and } j \tag{5}$$

$$\text{AnnualCap}_s = \sum_r \text{loadcap}_s W_{s,r} \frac{365}{\text{time}_r} \quad \forall\, s \tag{6}$$

$$\sum_r \sum_j Q_{s,r,j} \leq \text{AnnualCap}_s \quad \forall\, s \tag{7}$$

$$Q_{s,r,j} \geq 0 \quad \forall \text{ combinations of } s, r \text{ and } j \tag{8}$$

$$\text{AnnualCap}_s \geq 0 \quad \forall\, s \tag{9}$$